

LYNN LAKE, MB, Canada

WMO: 710780

Lat: 56.860N

Lon: 101.080W

Elev: 357

StdP: 97.11

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.1	-35.9	-42.6	0.1	-37.9	-40.4	0.1	-35.7	9.5	-20.0	8.5	-21.5	2.2	300	0.738

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	27.5	16.8	25.8	15.9	24.0	15.1	18.4	24.6	17.3	23.0	16.3	21.5	4.5	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.4	12.2	20.0	15.3	11.3	18.9	14.2	10.5	18.3	53.8	25.0	50.3	23.1	47.1	21.4	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
9.1	8.0	7.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-41.8	31.0	2.3	1.5	-43.4	32.1	-44.8	33.0	-46.1	33.9	-47.8	35.0
			-41.8	20.5	2.3	1.1	-43.4	21.2	-44.7	21.8	-46.0	22.4	-47.6	23.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.6	-23.3	-20.4	-13.2	-3.2	5.6	13.4	16.6	14.8	8.6	0.1	-11.4	-20.0
	DBStd	15.39	7.88	7.41	8.21	7.43	5.59	4.38	3.38	3.58	4.60	4.64	7.11	8.30
	HDD10.0	5136	1032	851	718	396	158	18	1	3	79	306	643	931
	HDD18.3	7676	1290	1084	977	645	395	154	73	120	292	564	893	1189
	CDD10.0	542	0	0	0	1	22	121	207	152	38	1	0	0
	CDD18.3	39	0	0	0	0	1	8	21	10	1	0	0	0
	CDH23.3	506	0	0	0	0	20	117	236	119	14	0	0	0
	CDH26.7	88	0	0	0	0	4	18	42	22	2	0	0	0
	Wind	WSAvg	3.2	3.2	3.3	3.4	3.4	3.5	3.4	3.1	3.2	3.3	3.0	2.9
(15) Precipitation	PrecAvg	535	24	18	18	26	46	67	80	77	64	47	35	25
	PrecMax	699	51	32	42	73	119	163	132	159	168	98	92	51
	PrecMin	335	9	1	5	2	5	14	45	25	12	16	9	10
	PrecStd	91	12	9	9	18	26	31	28	34	42	19	22	10
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.2	0.3	8.5	17.8	25.6	29.0	30.0	29.1	25.2	15.4	4.9	-0.8
		MCWB	-1.8	-0.9	3.6	9.3	13.6	17.0	18.1	18.1	16.3	9.4	2.8	-1.6
	2%	DB	-4.0	-2.1	4.8	13.2	22.3	26.5	27.7	26.4	21.2	11.5	1.8	-3.7
		MCWB	-4.6	-3.0	1.1	6.6	12.0	15.3	17.2	17.0	14.1	7.6	0.4	-4.3
	5%	DB	-7.7	-5.6	1.9	9.9	18.9	24.4	26.0	24.3	18.6	8.9	-0.1	-6.1
		MCWB	-8.1	-6.4	-0.4	4.5	10.2	14.4	16.2	16.0	13.0	6.5	-0.8	-6.5
	10%	DB	-11.7	-8.5	-0.7	7.2	15.8	22.2	24.2	22.0	15.9	6.6	-1.9	-8.0
		MCWB	-12.1	-9.2	-2.4	2.9	9.0	13.5	15.6	15.1	11.7	5.0	-2.5	-8.4
	0.4%	WB	-1.7	-0.8	3.8	9.9	14.7	19.1	20.2	19.5	17.3	10.2	2.8	-1.4
		MCDB	-1.4	-0.1	8.0	15.3	23.8	25.6	26.1	26.1	23.1	13.3	4.5	-0.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-4.6	-3.0	1.5	6.9	12.6	17.2	18.6	18.1	15.2	8.1	0.6	-4.3
		MCDB	-4.0	-2.1	4.3	12.7	19.7	22.5	24.8	24.5	19.6	11.0	1.5	-3.7
	5%	WB	-8.1	-6.4	-0.1	4.8	11.1	15.6	17.6	17.0	13.7	6.6	-0.8	-6.4
		MCDB	-7.7	-5.6	2.0	9.2	18.1	21.9	23.3	22.5	17.7	8.7	-0.1	-6.1
	10%	WB	-12.1	-9.2	-2.5	3.1	9.3	14.3	16.6	15.9	12.1	5.0	-2.5	-8.4
		MCDB	-11.8	-8.5	-0.7	7.0	15.1	20.9	21.9	20.2	15.5	6.6	-1.9	-8.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	11.1	12.7	12.2	12.0	11.6	11.1	10.2	9.0	6.3	7.5	8.2
		MCDBR	11.3	11.8	13.5	14.1	17.1	15.1	14.5	14.1	13.5	10.4	7.7	8.2
	5% WB	MCWBR	11.0	10.9	10.5	8.8	8.9	6.4	6.0	6.3	7.4	6.9	6.6	7.9
		MCDWR	11.2	11.5	12.6	13.2	15.2	12.7	12.1	12.2	11.9	9.5	7.3	8.2
		MCWBR	10.9	10.6	9.9	8.5	8.3	6.2	5.6	6.3	7.1	6.6	6.3	7.9
(40) Clear-Sky Solar Irradiance	taub		0.223	0.248	0.266	0.313	0.327	0.352	0.387	0.359	0.311	0.284	0.232	0.210
	taud		2.233	2.279	2.287	2.303	2.467	2.441	2.368	2.464	2.573	2.559	2.358	2.158
	Ebn at Noon		701	832	905	904	913	889	846	850	854	792	694	602
	Edh at Noon		48	72	94	110	100	104	110	94	73	56	43	39
(44) All-Sky Solar Radiation	RadAvg		0.72	1.78	3.42	5.11	5.70	5.75	5.33	4.26	2.83	1.37	0.63	0.43
	RadStd		0.07	0.12	0.24	0.34	0.52	0.53	0.39	0.35	0.36	0.17	0.06	0.06

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page